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NF270.5 data sets

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To: Xinhong Liu <xliu27@nd.edu>

Cc: Jonathan Ouimet <jouimet@nd.edu>

Thank you for sending this! I noticed that the data you have is different from my values. The reason for this is that we ran into some issues with our ICP analysis. The instrument was not giving us reliable data, later to find out there was contamination within the apparatus which was skewing our results.

After fixing the issue, we re-ran every sample and therefore I updated the experimental data according to the most recent analysis (the conductivity probe calibration curve remains the same, however). I have attached EXCEL sheets and raw data for all the requested NF270.5 experiments as well as the duplicates (NF270.6) and triplicates (NF270.7). Please let me know if I missed one. I know this is a lot of information within one email, but I figured it's easier to have everything in one place.

Another thing to mention is that ICP reads the diafiltrate concentration higher than the probe does. We ran some additional experiments to elucidate why we saw this behavior. We were able to verify that the error was coming from the solution dilution during ICP sample preparation. Additionally, the probe calibration curve is only calibrated to 100mM, however, we are running experiments at 120mM. Error within the extrapolation could also be creeping into the probe readings.

Best,

Laurianne

[Quoted text hidden]

19 attachments

-  **Exp_01302023_NF270_7_lag_D_SS_KCl.txt**
20K
-  **Membrane.NF270.5.xlsx**
127K
-  **Membrane.NF270.7.xlsx**
35K
-  **Membrane.NF270.6.xlsx**
36K
-  **Exp_01312023_NF270_7_D_SS_KCl.txt**
15K
-  **Exp_01302023_NF270_6_lag_D_SS_KCl.txt**
17K
-  **Exp_02012023_NF270_6_Lag_D_SS_KCl.txt**
14K
-  **Exp_10262022_NF270_5_lag_D_SS_KCl.txt**
15K
-  **Exp_10252022_NF270_5_Lag_D_SS_KCl.txt**
15K
-  **Exp_11022022_NF270_5Rev_D_SS_KCl.txt**
15K
-  **Exp_11032022_NF270_5RevB_D_SS_KCl.txt**
19K
-  **Exp_11022022_NF270_5Overflow_D_SS_KCl.txt**
41K

-  **Exp_11022022_NF270_5_Lag_F_SS_KCl.txt**
25K
-  **Exp_11022022_NF270_5LagB_D_SS_KCl.txt**
26K
-  **Exp_11142022_NF270_5A_D_SS_KCl.txt**
14K
-  **Exp_11032022_NF270_5B_D_SS_KCl.txt**
21K
-  **Exp_01302023_NF270_5_Bover_D_SS_KCl.txt**
21K
-  **Exp_08222023_NF270_5_local_D_SS_KCl.txt**
19K
-  **Exp_08222023_NF270_5_lowP_D_SS_KCl.txt**
56K